A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 10

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 3 points below. **1** Negative feedback responds to any change by re-establishing the previous equilibrium **1** Only large unusual events overwhelm the feedback so a new equilibrium forms **1 Example:** methane hydrate release at the end of the Permian causing climate change and mass extinction **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any appropriate named example, including ones that are not listed here. • **negative feedback** → a change that cancels itself out / a self-correcting cycle	3

Do not write outside the box

Question	Marking guidance	Marks
01.2	**1 mark each** 1 mark for each of the 4 points below. **1** In a diverse system each component is relatively less important so the loss / change of one has a small effect – alternative food sources and relationships buffer the change **1 Natural:** coral reefs and tropical rainforests with complex food webs are stable – simple polar / desert systems fluctuate **1 Human:** agroecosystems focus on a few crops and remove detritivores, pollinators and predators so are vulnerable to pests, disease, weather **1 Diverse technical systems:** a society using a range of renewable energy resources is more secure than one relying on a single source **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts	4
01.3	**Any 3 of 7** 1 mark each for any 3 of the 7 points below, in any order, maximum 3. **1** Melting permafrost releasing methane **1** Reduced albedo as ice / snow melt **1** Methane hydrate release **1** Forest and peat fires **1** Faster soil decomposition **1** Ocean acidification reducing sequestration **1** Cirrus cloud formation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **positive feedback** → a change that makes itself larger / a self-reinforcing cycle • **albedo** → reflectivity / the proportion of solar radiation reflected / how much sunlight is bounced back **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • carbon dioxide as the main cause of acid rain - the gases responsible are sulfur dioxide and oxides of nitrogen • 'ice reflects the heat back' - albedo is the reflection of incoming short-wave solar radiation, before it is absorbed	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 371-372; AQA spec 3.6.1.	

Do not write outside the box

Question	Marking guidance	Marks
02.1	**1 mark each** 1 mark for each of the 3 points below. **1** As sea otters declined (1960-1980) sea urchins increased (to 350) and kelp area fell (to 25) **1** As otters recovered after 1990 urchins declined and kelp recovered **1** Otter and kelp trends are positively correlated – urchins negatively correlated with both **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
02.2	**1 mark each** 1 mark for each of the 3 points below. **1** Sea otters are predators of sea urchins so fewer otters allowed the urchin population to rise **1** Sea urchins graze kelp so the kelp forest was over-grazed and destroyed **1** Kelp forests are habitat / food for many species including otters (keystone species / trophic cascade) – otter recovery restored the balance **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **keystone species** → a species whose loss changes the whole community **Do not credit** • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	3
02.3	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Over-fishing of sandeels reduced puffins **1** Over-collection of turtle eggs reduced turtles so jellyfish (their food) increased – over-fishing of sardines / herring (competitors) also increased jellyfish **1** Loss of pollinating bees (pesticides, wild flower loss) reduces plant reproduction – loss of elephants / large herbivores reduces seed dispersal of trees **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	2

Do not write outside the box

Question	Marking guidance	Marks
02.4	**1 mark each** 1 mark for each of the 2 points below. **1 They control abiotic features:** beaver dams create ponds colonised by aquatic species – elephants create clearings and keep water holes open **1** Species that depend on these features (light, water, minerals) cannot survive in their absence **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **abiotic** → non-living / physical factors	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 23-25; AQA spec 3.1.2.2.	
03.1	**2 marks per point** 2 marks for each of the 4 points below; award 1 mark where a point is only partly made. **2 Species:** a group of closely related organisms that naturally interbreed to produce fertile offspring and are reproductively isolated from other groups (Passer domesticus, house sparrow) – the only 'real' taxon **2 Niche:** the role a species plays in its habitat - how it uses resources and relates to other species (tawny owl: nocturnal carnivore nesting in tree holes eating small mammals) – two species cannot share exactly the same niche in one habitat **2 Community:** all the populations of all species (plants, animals, fungi, bacteria, protists, archaea) living in an area **2 Biome:** a large geographical region with a specific climate and characteristic community, including all areas where it occurs (all tropical rainforests, tundra, savannah) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any appropriate named example, including ones that are not listed here. • **Niche** → the role a species plays and the conditions it needs	8
03.2	**1 mark each** 1 mark for each of the 2 points below. **1** An ecosystem is the biotic community plus abiotic factors of one area, relatively self-contained (a particular coral reef) **1** A biome includes all separate ecosystems with the same climate and community type (all coral reefs) – each reef is a different ecosystem within the same biome **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **biotic** → living factors / effects of other organisms • **abiotic** → non-living / physical factors	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 82-85; AQA spec 3.1.3.2.	

Do not write outside the box

Question	Marking guidance	Marks
04.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** Extensive cereal 5.0 – intensive cereal 3.0 **1** Intensive beef 0.1 **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. **Do not credit** • A bare wrong answer with no working.	2
04.2	**1 mark each** 1 mark for each of the 3 points below. **1** Intensive cereal is most productive per hectare (60 GJ) but less efficient (3.0) than extensive (5.0) **1** Each extra unit of input gives a smaller extra yield (law of diminishing returns) **1** Beef has very low productivity and efficiency because energy is lost at an extra trophic level **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **trophic level** → the feeding level in a food chain **Do not credit** • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	3
04.3	**Marks shared** Up to 3 marks in total - share them between the 2 points below according to how completely each one is made. **up to 2** Any input that increases productivity but requires the use of energy (usually fossil fuel) **up to 2** Manufacture of nitrate fertiliser (Haber process) and pesticides – fuel for machinery – pumping irrigation water – grain drying – food processing and transport – embodied energy of machinery **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **embodied energy** → all the energy used to make and transport a product **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	3

Do not write outside the box

Question	Marking guidance	Marks
04.4	**1 mark each** 1 mark for each of the 2 points below. **1** Because of diminishing returns, the first units of input on unfertilised land give the largest yield increases **1** Moving inputs from intensively farmed land (where extra input adds little) to extensively farmed land gives a greater total yield - though land, cost and distribution limit this **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 324-327; AQA spec 3.5.1.2.	
05.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B	1
05.2	**1 mark each** 1 mark for each of the 2 points below. **1** Number all possible sampling sites (eg on a grid map) **1** Select the sites using a random number generator / table – (regular shape: grid coordinates from random numbers) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **random** → positions chosen using random numbers so the choice is not biased **Do not credit** • 'pick places at random by eye' - positions have to be generated with random numbers to remove bias	2
05.3	**1 mark each** 1 mark for each of the 2 points below. **1** Sites may be chosen for convenience or because they appear to support (or dismiss) the hypothesis **1** So the sample does not represent the whole area and conclusions are invalid **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 391-393; AQA spec 3.7.1.	

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Question	Marking guidance	Marks
06.1	**1 mark each** 1 mark for each of the 3 points below. **1** Small clearings were farmed for a few years then abandoned as nutrients declined – the forest recolonised and soil fertility recovered before the area was cleared again **1** Population growth shortens the gap between clearances so the forest and soil do not recover **1** Clearings enlarge and degrade – less regrowth **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	3
06.2	**1 mark each** 1 mark for each of the 2 points below. **1** Rainforest soils are thin and nutrient-poor (nutrients held in the biomass) – once cleared, fertility falls in a few years and yields drop, so more land is cleared **1** Abandoned land may not be recolonised if no forest survives nearby – erosion of exposed soil **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
06.3	**1 mark each** 1 mark for each of the 3 points below. **1** Mining (aluminium, iron, gold, copper) destroys habitat directly and pollutes rivers with turbid drainage, cyanide and mercury from processing **1** Reservoirs (eg Tucurui, 2850 km^2) flood large areas, block migratory fish and river dolphins – decomposing vegetation produces acidic, deoxygenated water – fluctuating downstream flows **1** Droughts increase frequency, area and duration of fires (long fires boil sap and kill trees) – cloud forests lose humidity as clouds form higher so frogs and epiphytes decline – some models predict replacement by savannah **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	3

Do not write outside the box

Question	Marking guidance	Marks
06.4	**1 mark each** 1 mark for each of the 2 points below. **1** A creditor country or conservation charity cancels / buys part of a debtor country's foreign debt **1** In return the country guarantees to protect an area of rainforest (eg USA - Guatemala, Brazil; WWF - Costa Rica, Philippines) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 55-57; AQA spec 3.1.2.3.4.	
07.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B - radio waves do not travel through seawater – acoustic or satellite tags transmit position	1
07.2	**2 marks per point** 2 marks for each of the 2 points below; award 1 mark where a point is only partly made. **2** Shows survival, dispersal and breeding of released individuals so success can be assessed and future releases improved **2** Radio / GPS tags or collars – leg rings / wing tags on birds – camera traps – DNA from droppings to confirm identity and diet **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **captive breeding** → breeding in captivity to increase numbers for release **Do not credit** • captive breeding alone as conservation - credit needs reintroduction, habitat protection or genetic management as well	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 393-394, 402-407; AQA spec 3.1.2.3.5-3.1.2.3.6.	
08.1	**1 mark each** 1 mark for each of the 4 points below. **1 Energy:** low energy-density renewable (solar) / non-renewable fossil fuels **1 Materials:** cyclic - wastes of one process are raw materials of the next / linear - used once then discarded, depletion **1 Wastes:** biodegradable, non-toxic or do not build up / toxic, persistent (eg PCBs, organochlorines, toxic metals) that bioaccumulate **1 Temperature:** low / high (Haber process, incineration) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **biodegradable** → broken down by decomposers • **persistent** → does not break down / stays in the environment for a long time • **incineration** → burning waste, usually with energy recovery **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts	4

Do not write outside the box

Question	Marking guidance	Marks
08.2	**1 mark each** 1 mark for each of the 2 points below. **1** Abundant simple raw materials build into monomers and then a wide range of polymers (carbohydrates, proteins) **1** They are re-used in cycles so there is no depletion of raw materials or build-up of wastes **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
08.3	**2 marks** 2 marks for the point below; award fewer where it is only partly made. **2** Use often disperses the minerals (eg in products spread over the world / corrosion) **OR** produces mixtures of materials that are difficult to separate, so sustainable exploitation is reduced **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	2
08.4	**1 mark each** 1 mark for each of the 2 points below. **1** They cannot easily be removed from end-of-life appliances and are not biodegradable **1** Natural wastes are non-toxic / broken down to raw materials for other processes, whereas persistent toxins may bioaccumulate and biomagnify **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodegradable** → broken down by decomposers • **persistent** → does not break down / stays in the environment for a long time	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373-374; AQA spec 3.6.2-3.6.3.	

Do not write outside the box

Question	Marking guidance	Marks
09.1	**1 mark each** 1 mark for each of the 6 points below. **1** Cycling of materials (like biogeochemical cycles) - recycling, re-use **1** Energy from renewable sources **1** Human activities should support ecosystems / conserve biodiversity **1** Separation of technical and biological materials – connected systems where the waste of one process feeds another (industrial symbiosis) **1** Diverse systems are more resistant to change **1** Design of products for extended use and end-of-life re-use – optimum rather than maximum production – technologies that improve efficiency and enable dismantling **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **recycling** → reprocessing a material to make something new • **biodiversity** → the variety of life: habitat, species and genetic diversity • **symbiosis** → a close relationship in which both species benefit **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • 'recycling uses no energy' - collection, sorting and reprocessing all use energy, usually less than making the material from new	6
09.2	**1 mark each** 1 mark for each of the 2 points below. **1** Natural systems over-produce so the surplus supports the processes the system relies on: plant products feed pollinators, seed dispersers, decomposers, mycorrhizae **1** Agroecosystems aim for maximum harvested yield, removing the surplus that would maintain soil biota, pollinators and pest predators, so they need artificial inputs **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	2
09.3	**1 mark each** 1 mark for each of the 2 points below. **1** Costs of environmental measures could put companies in one country at a disadvantage against competitors in countries without them **1** A common approach gives a level playing field, coordinated monitoring and support for less successful countries **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 374-375; AQA spec 3.6.4.1.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**1 mark each** 1 mark for each of the 2 points below. **1** Plants produce chemicals (often alkaloids) that are toxic to the herbivores that would eat them **1** In carefully controlled amounts these chemicals have beneficial physiological effects in humans **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
10.2	**1 mark each** 1 mark for each of the 2 points below. **1** Only a very small proportion of species have been studied for medicinal substances, so it is not known which species will be useful **1** Conserving the habitat protects all the species in it, including those not yet studied and the species they depend on **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
10.3	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Marsupials (kangaroo / wallaby): young develop in the pouch so are easier to study than a human foetus - developmental problems **1 Hippopotamus:** hipposudoric acid skin secretion (sunscreen / antimicrobial) - burns treatment **1 Marine sponges:** proteins preventing graft rejection - organ transplants **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any appropriate named example, including ones that are not listed here.	2

Do not write outside the box

Question	Marking guidance	Marks
10.4	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Resources:** wood, fibres, oils, fuels, food – new food species adapted to local conditions (morama bean, yeheb, perennial maize, Kernza, bison, eland) • **Biomimetics:** whale flipper tubercles, shark skin, termite-mound ventilation, bird bones, gecko adhesion, burrs / Velcro, lotus self-cleaning, spider silk • Medicines and physiological research – genetic resources - crop wild relatives, Vavilov centres, gene pools, inbreeding • **Ecosystem services:** atmospheric composition, hydrological cycle, biogeochemical cycles, soil maintenance – interspecies relationships - food, pollination, seed dispersal, habitat provision • **Evaluation:** services go unnoticed because of dynamic equilibrium – unknown future value – economic vs moral / intrinsic arguments – conservation also needed for aesthetic and cultural reasons • Justified conclusion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **biodiversity** → the variety of life: habitat, species and genetic diversity	9
	Model answer - Level 3 (9 marks)	
	Other species supply resources we use directly: wood, fibres such as cotton and wool, oils, fuels and food. Because most farmed species were domesticated long ago and are often kept where they are poorly adapted, wild species such as the drought-resistant morama bean, the yeheb tree that grows on poor soils, perennial maize that needs no ploughing, and bison that graze grassland too poor for cattle offer new, better-adapted food sources. Wild species also carry genetic resources: crop wild relatives in Vavilov centres hold genes for disease resistance and salt tolerance that inbred commercial varieties lack.	

Do not write outside the box

Question	Marking guidance	Marks
	Biomimetics copies adaptations into technology - humpback flipper tubercles in turbine blades, shark skin in friction-reducing coatings, termite-mound convection in naturally ventilated buildings, hollow bird bones in lightweight roofs. Medicines such as taxol, quinine, AZT and diosgenin came from wild species, and physiological research on marsupials, hippo skin and sponge proteins is helping with foetal development, burns and transplant rejection. Since only a tiny fraction of species has been studied, the unknown future value argues for conserving whole habitats.	
	Beyond resources, other species provide services we could not replace: regulation of atmospheric gases, evapotranspiration driving rainfall, the carbon, nitrogen and phosphorus cycles run by microbes, soil formation by decomposers, and pollination, seed dispersal and habitat provision. These are unnoticed precisely because they are in dynamic equilibrium, so their economic value is underestimated. On balance the resource and service arguments alone justify conservation, but they are strongest when combined with the intrinsic and aesthetic case, because many species with no known use still sustain the systems we depend on.	
	Weaker answer - Level 1 (3 marks)	
	Biodiversity is important because we get lots of things from other species. We get wood, cotton and food from plants and animals. Some medicines come from plants, like aspirin from willow trees. Animals can help design things, like Velcro was copied from seeds. Other species also do things for us like bees pollinating crops. If we lose species we might lose things that could be useful in the future. So we should conserve biodiversity.	
	What makes the difference	
	• Both answers cover resources, medicines, biomimetics and services, but Level 3 develops each with specific, accurate detail (which species, what property, why it matters).	
	• The question asks 'to what extent'. Level 3 weighs resource arguments against intrinsic value and reaches a reasoned conclusion; Level 1 asserts 'so we should conserve'.	
	• Level 3 explains *why* services are undervalued (dynamic equilibrium) - an explanation, not just a list.	
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 10-19; AQA spec 3.1.2.1.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Discuss the causes and consequences of deforestation and evaluate the extent to which forests can be exploited sustainably.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Importance:** resources (timber, fuel, food, fibres, medicines) and ecosystem services (atmospheric regulation, carbon store, hydrological regulation, climate, soil conservation, habitat, recreation) • **Causes:** agriculture (subsistence and cash crops), mineral extraction, reservoirs, urbanisation, transport, unsustainable timber, fuel and fodder exploitation above MSY – trends by region • **Consequences:** loss of resources, biodiversity and species, fragmentation – hydrological changes – soil impacts – climate impacts (albedo, carbon, rainfall) • **Productivity vs biodiversity:** non-indigenous species, monocultures, close planting, simple age structure, clear-felling • **Sustainable management:** MSY harvest rates, mixed indigenous species, mixed age structure, selective logging, traditional coppicing / pollarding, FSC certification, community involvement, protected areas, debt-for-nature, afforestation programmes (China) – Rio Bravo case study	25
	• **Evaluation:** economics of slow-growing hardwoods, enforcement, demand from MEDCs, ITTO failures – judgement *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 50-57, 360-369, 386-387; AQA spec 3.5.3, 3.1.2.3.4.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Using named examples, discuss how the ecological features of a habitat determine its importance, the threats it faces and the conservation strategies that are appropriate.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Temperate broadleaf woodland:** seasonality, deep soils, high biodiversity, resources, climate and soil regulation – deforestation, fragmentation, management change – SSSIs, ancient woodland protection, coppicing, mixed planting • **Tropical rainforest:** stable warm wet climate, stratification, very high biodiversity, carbon and hydrological roles, thin soils – fuelwood, timber, agriculture, mining, reservoirs, climate change – protected areas, debt-for-nature swaps, sustainable exploitation • **Tropical coral reefs:** cnidarian-zooxanthellae symbiosis, light, temperature, turbidity, salinity – fisheries, erosion protection, medicines, tourism – damage, souvenirs, sedimentation, pollution, bleaching, fishing, introduced species – MPAs, no-take zones, control of activities • **Deep-water reefs:** cold, dark, slow growth – research, fisheries – trawling, oil / gas, acidification – SACs • **Oceanic islands:** isolation, endemism, no mammal predators	25

Do not write outside the box

Question	Marking guidance	Marks
	– exploitation, introduced species, habitat change, sea level – eradications, control of development and visitors • **Mangroves:** halophytes, anaerobic mud – erosion protection, fisheries nurseries, sediment trapping – clearance for aquaculture / urban / ports, pollution, climate change – protection, replanting • **Antarctica:** extreme cold, seasonality, marine nutrients, ice – water store, albedo, carbon, research – climate change, ozone, tourism, fishing, minerals – Antarctic Treaty, CCAMLR, tourism and waste rules • **Synthesis:** features -> importance -> threats -> strategy – evaluation of success – conclusion *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 50-71; AQA spec 3.1.2.3.4.	

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